the points of resonance in the $_{92}U^{238}$, and so a greater number of neutrons are captured by these nuclei. The heating thus atomatically brakes down the chain reaction, so that the entire apparatus becomes stabilized at a certain temperature, the magnitude of which depends on its size and geometric design. Furthermore, it is possible to introduce, into the reactor, from outside, some substance that absorbs neutrons (cadmium is the most suitable material for this purpose) and consequently will act as an added moderator of the chain reaction and regulate the temperature artificially. Once a certain temperature is reached, the reactor will maintain it quite independently of the quantity of energy removed from it. If a considerable amount of energy has been removed—in consequence for instance of good thermal conductivity—the reactor cools off a little at once. The frequency of the disintegration processes immediately increases enormously, and the original temperature is re-established.

Figure 40A shows the interior of the model of a uranium reactor, composed of uranium and heavy water. It was installed, during the war, in a cellar carved in natural rock, in the village of Haigerloch in Württemberg, by a working team of the *Kaiser Wilhelm Institut* (*Wirtz, Bopp, Fischer, Jensen, Ritter*). The photograph shows a great many disks of metallic uranium, suspended by chains from a lid which can be lowered into a tank containing heavy water. The tank itself is enclosed in a thick layer of graphite, which, however, is hardly visible in this picture. Figure 40B is a schematic digram of the ura reactor; the shaded area is the graphite coating. The apparatus stood in a large water tank. A neutron sou hanging in the centre, for observation; measuring leads (attached near the exterior. The apparatus was still a litt small to sustain a fission reaction independently, but a increase in its size would have been sufficient to start o process of energy production.